FORM REF-HS · REV. 5 POST-STASIS REFRESHER · REFERENCE SHEET POST IN EVERY CRANE CAB, LIFT MUSTER AND EQUIPMENT YARD

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

HAND SIGNALS.

IN FULL VIEW OF THE OPERATOR, YOUR ARMS DO THE TALKING. LOSE SIGHT, AND THE LIFT STOPS.

READ FIRST

ONE SIGNAL PERSON. ANYONE CAN STOP.

Each lift has **one** qualified signal person (orange hard hat, badge endorsement). The operator takes signals only from them, except **STOP and EMERGENCY STOP, which the operator obeys from anyone.** Signals are given facing the operator, in clear view. If the operator loses sight of the signal person or the load, the operator stops.

Radio drops during a lift? The operator stops all motion and holds the load. Resume only when the radio is back and tested, or when the signal person is in the operator's full, continuous view and the operator, signal person and whoever directs the lift (the Lift Director, or the lift lead on a routine lift) agree to switch to hand signals. Blind lifts stay stopped. Radio signals go on the lift talkgroup named on the lift plan, never on Ch 11 (REF-RAD, HS-06).

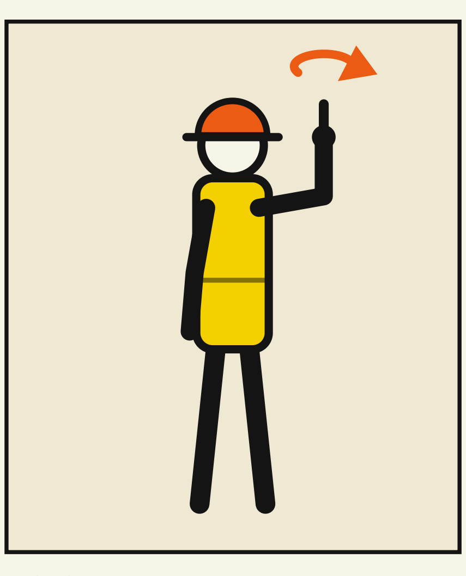
Site signals are for people, never for operators, except ALL STOP (the EMERGENCY STOP signal). Give EMERGENCY STOP first, then the site signal once the load is held.

At night, use lighted wands or lighted gloves. Sand, haze or dark so thick the operator can't clearly see every signal? Stop the lift.

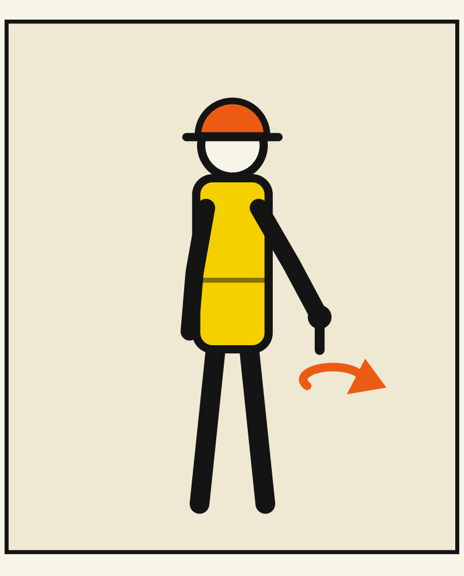
Figures face you, the way the operator sees the signal person. **Orange arrows** show the motion.

CRANE SIGNALS

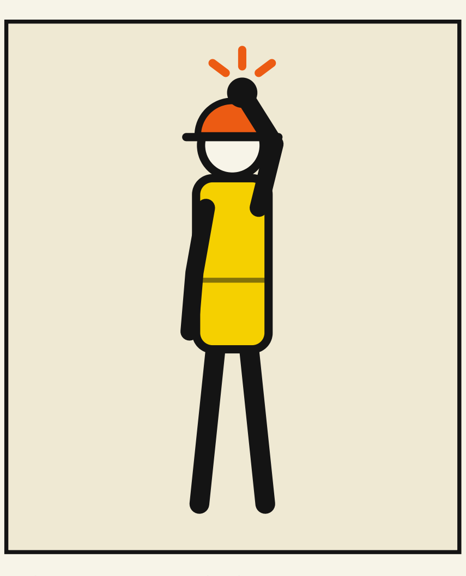
OLD EARTH STANDARD · ASME B30.5 MOBILE CRANES



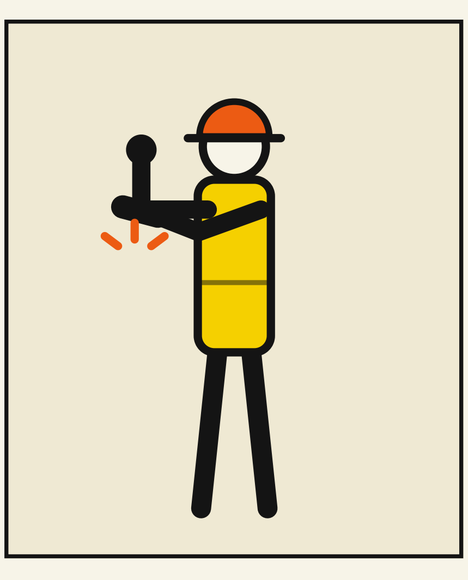
HOIST
Forearm vertical, forefinger pointing up. Move hand in small horizontal circles.



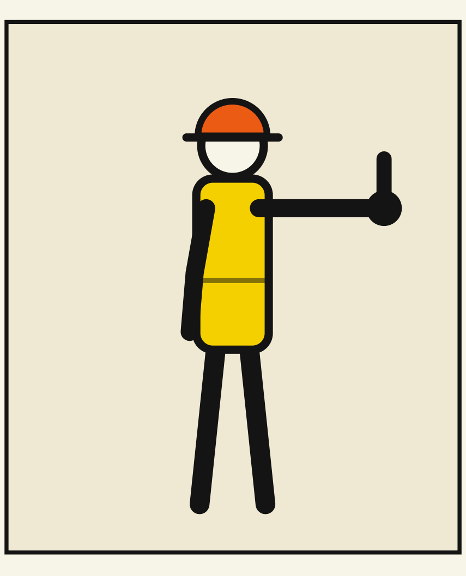
LOWER
Arm extended downward, forefinger pointing down. Move hand in small horizontal circles.



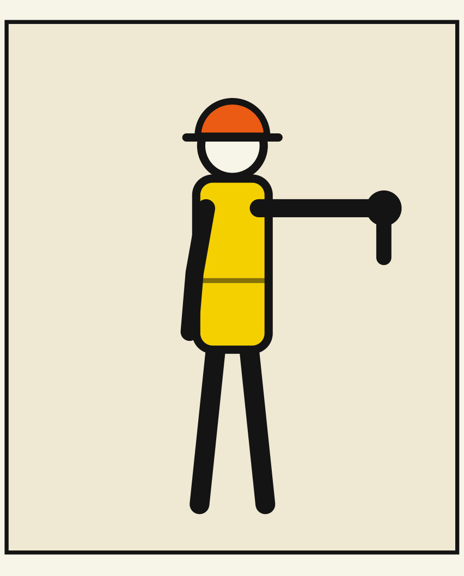
USE MAIN HOIST
Tap fist on top of head. Then give regular signals.



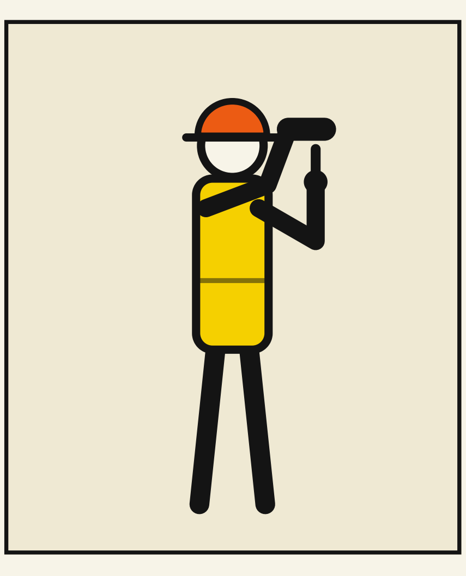
USE WHIPLINE
Tap the elbow with one hand (auxiliary hoist). Then give regular signals.



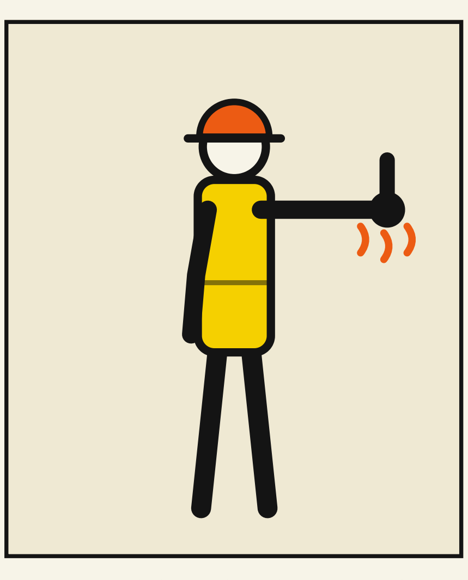
RAISE BOOM
Arm extended, fingers closed, thumb pointing upward.



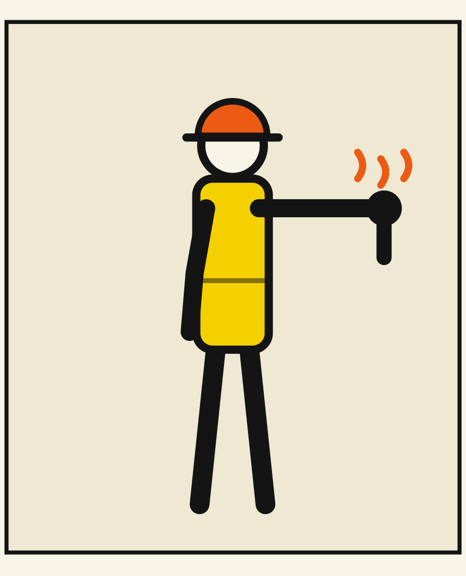
LOWER BOOM
Arm extended, fingers closed, thumb pointing downward.



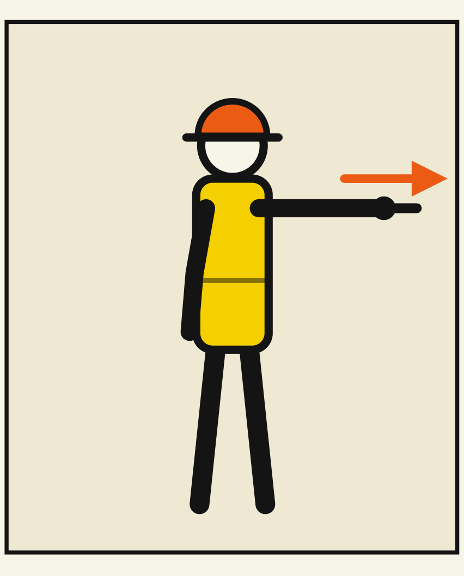
MOVE SLOWLY
One hand gives any motion signal. The other hand is held still, flat, in front of it. Shown: hoist slowly.



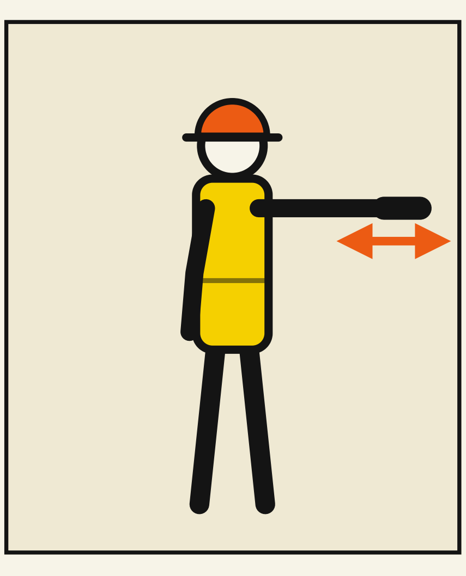
RAISE BOOM & LOWER LOAD
Arm extended, thumb up. Flex fingers in and out as long as load movement is wanted.



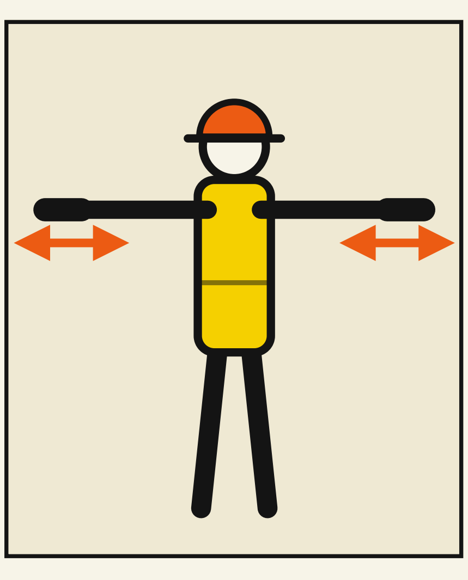
LOWER BOOM & RAISE LOAD
Arm extended, thumb down. Flex fingers in and out as long as load movement is wanted.



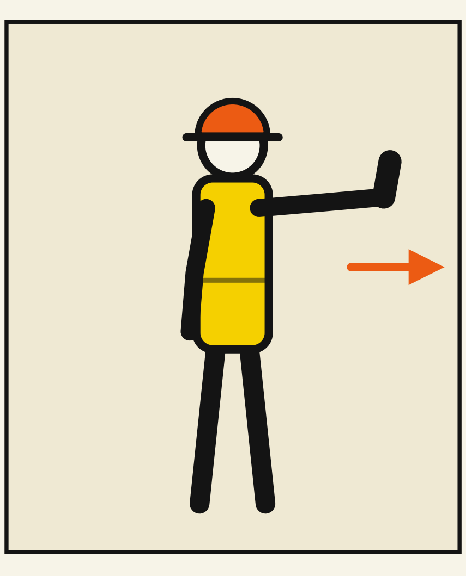
SWING
Arm extended, point with finger in the direction to swing the boom.



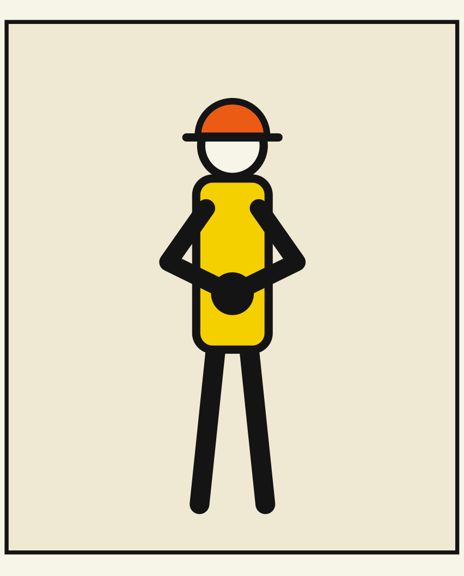
STOP
Arm extended, palm down. Move arm back and forth horizontally: swing it forward and back, level with your shoulder.



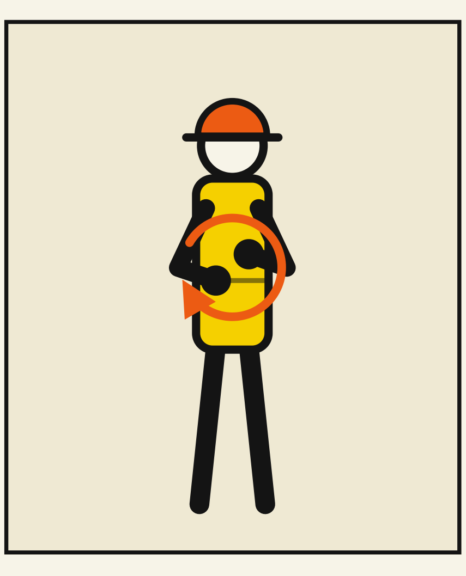
EMERGENCY STOP
Both arms extended, palms down. Move arms back and forth horizontally, level with your shoulders. Obeyed from anyone.



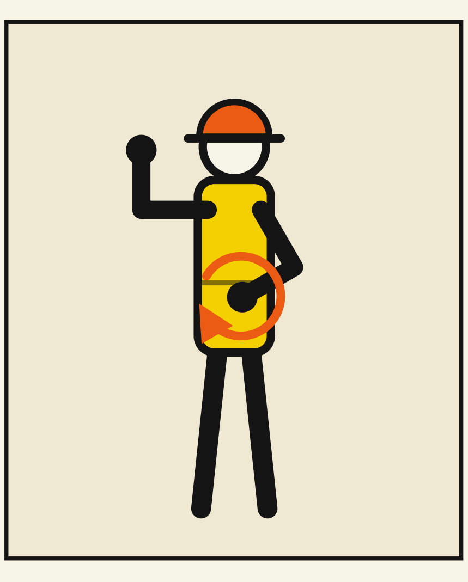
TRAVEL
Arm extended forward, hand open and slightly raised. Make a pushing motion in the direction of travel. (Arm drawn to the side so you can see it.)



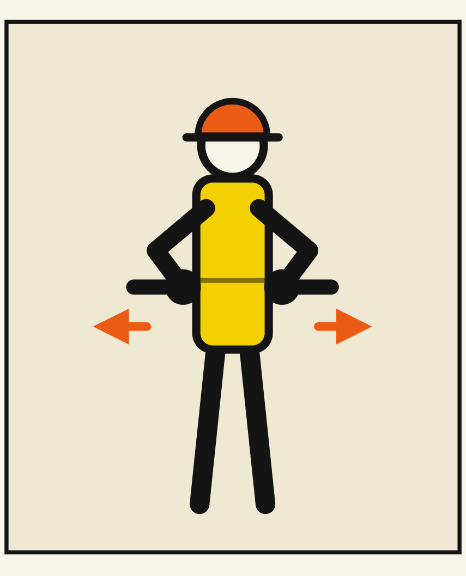
DOG EVERYTHING
Clasp hands in front of body. Stop all motion and hold everything where it is.



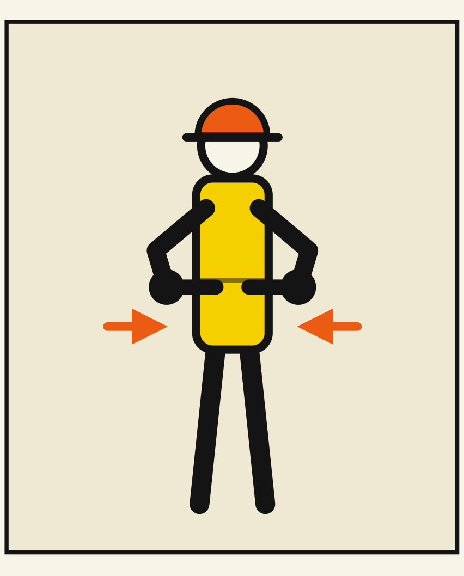
TRAVEL, BOTH TRACKS
Both fists in front of body, circling each other forward or back in the direction of travel. Crawler cranes only.



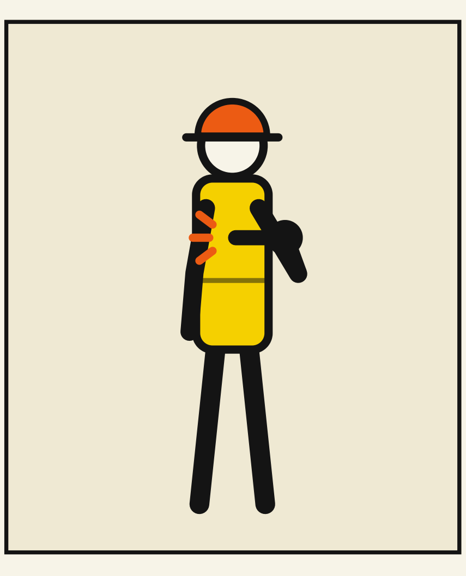
TRAVEL, ONE TRACK
Raised fist locks the track on that side. The other fist circles vertically in front of body, in the direction the other track travels. Crawler cranes only.



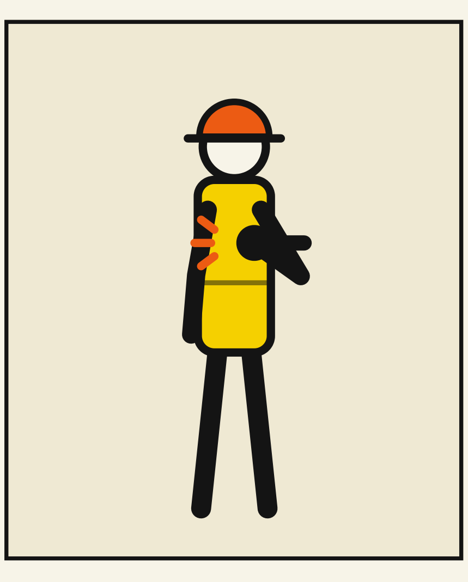
EXTEND BOOM
Both fists in front of body, thumbs pointing outward (telescoping boom).



RETRACT BOOM
Both fists in front of body, thumbs pointing toward each other (telescoping boom).



EXTEND BOOM, ONE HAND
One fist in front of chest, thumb tapping chest (telescoping boom).

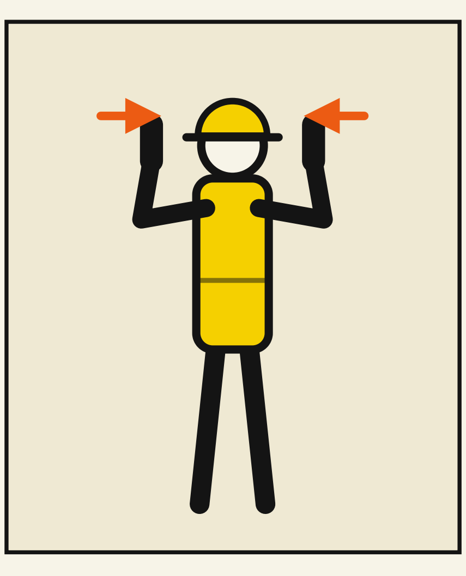


RETRACT BOOM, ONE HAND
One fist in front of chest, thumb pointing outward, heel of fist tapping chest (telescoping boom).

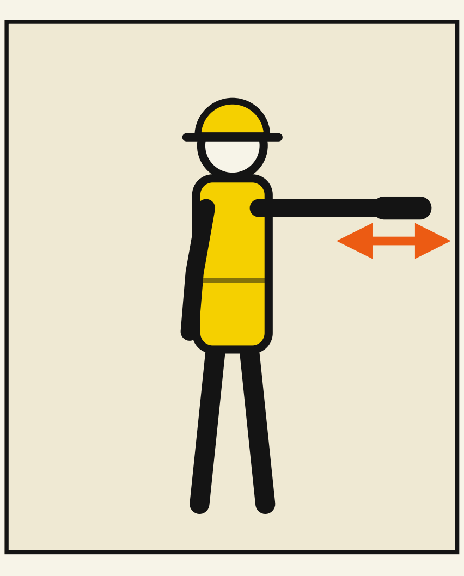
Crawler and telescoping-boom signals apply only to those cranes. Any signal not on this sheet is used only if the operator, signal person and whoever directs the lift (the Lift Director, or the lift lead on a routine lift) agree on it before the lift.

EQUIPMENT SPOTTER SIGNALS

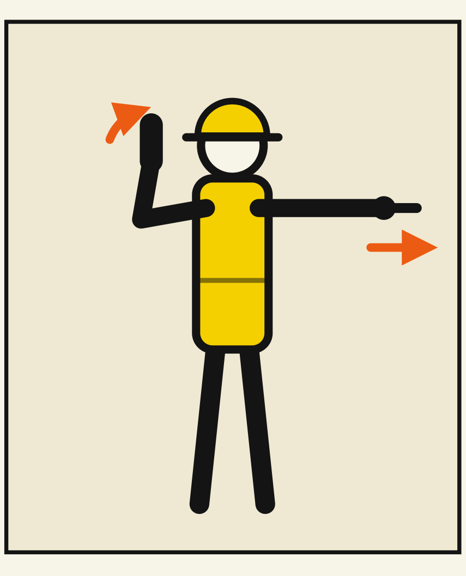
BACKING AND POSITIONING HAUL TRUCKS, LOADERS, CARRIERS



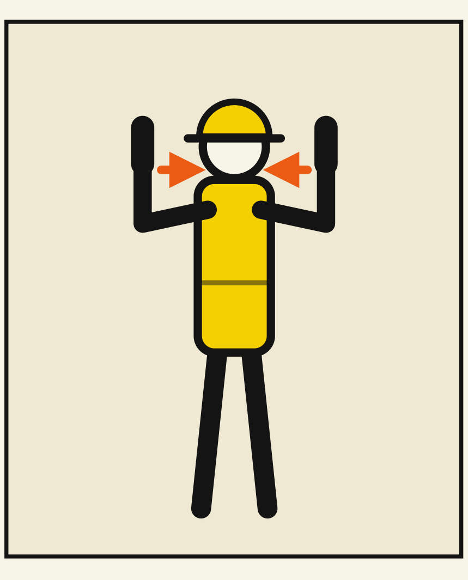
BACK UP / COME TO ME
Both hands up at head height, palms facing your own face. Pull your hands toward yourself again and again: keep coming back toward me.



STOP
Same as the crane STOP: arm out, palm down, swing it forward and back, level with your shoulder. Both arms = emergency stop. Hold it until the machine has stopped.



TURN
Point the direction to steer with one arm. Keep beckoning with the other hand.

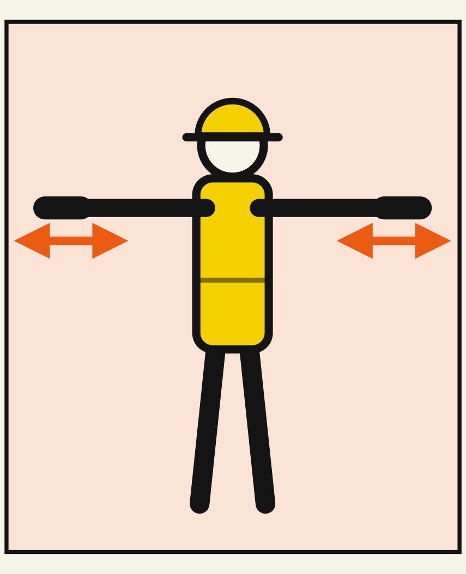


DISTANCE LEFT
Palms facing each other. The gap between hands is the distance left. Close them as it nears. Hands together = stop.

Spotters stand where the operator can see them in a mirror or camera, never directly behind the machine. Lose sight of your spotter: stop. Machine horns are about that machine only, never site signals: 1 = stopping, 2 = moving forward, 3 = backing up. Hear 3 horn blasts near you? Get out of its path.

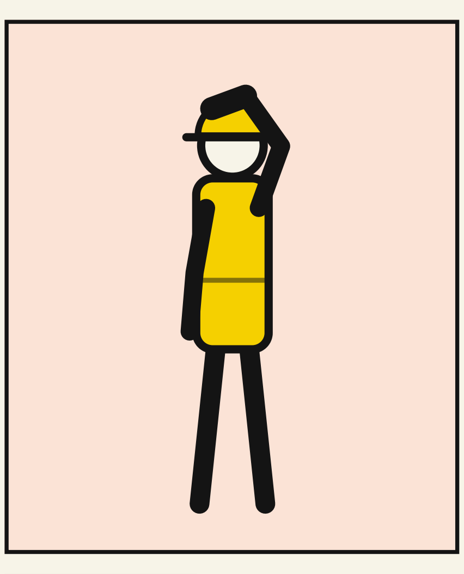
DCW SITE EMERGENCY SIGNALS

SITE RULE · DUNEWRIGHT ONLY, NOT AN OLD EARTH STANDARD



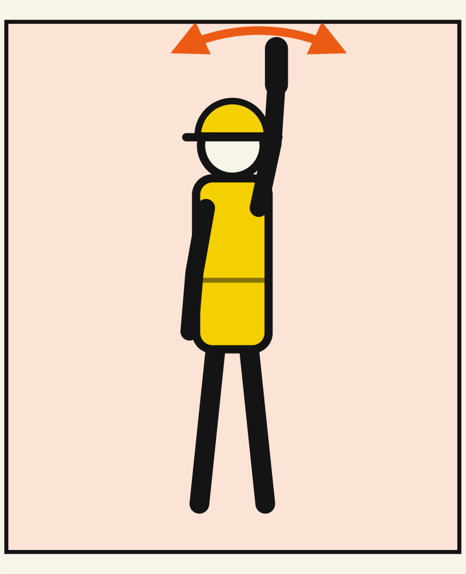
SITE SIGNAL

ALL STOP
Same as the crane EMERGENCY STOP, and the one site signal operators obey. Every machine, crane and torch in sight stops. Hold position.



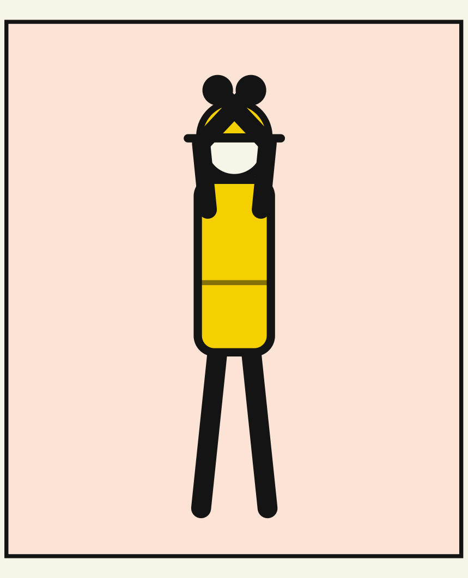
SITE SIGNAL

ARE YOU OK? / I'M OK
Arm arched over your head, open hand resting flat on top of your helmet. Hold it still, never tap. Ask with it, answer with the same signal. No answer? Go to them.



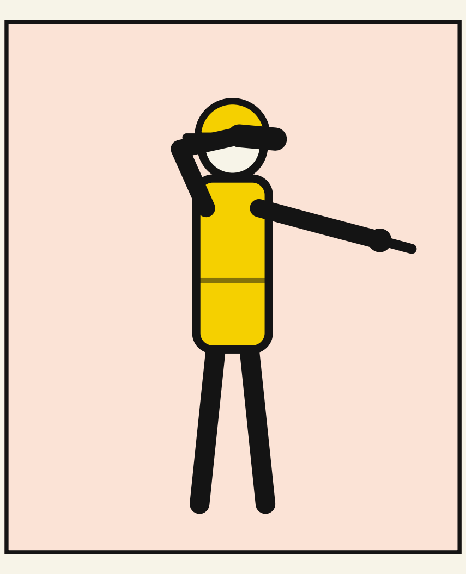
SITE SIGNAL

NEED HELP
Arm straight up, open hand, waved side to side overhead. Keep waving until someone reaches you.



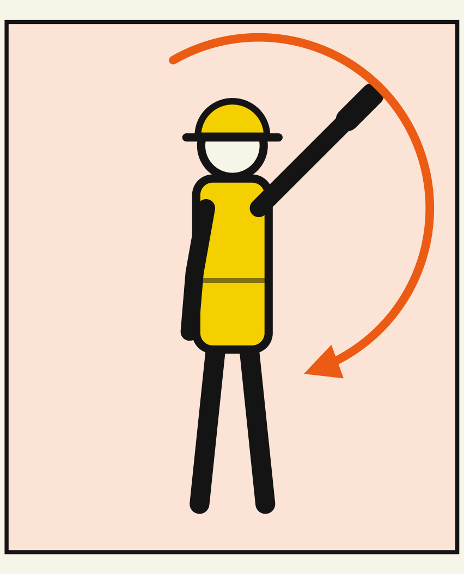
SITE SIGNAL

MAN DOWN
Cross both arms overhead in an X. Then point at the casualty.



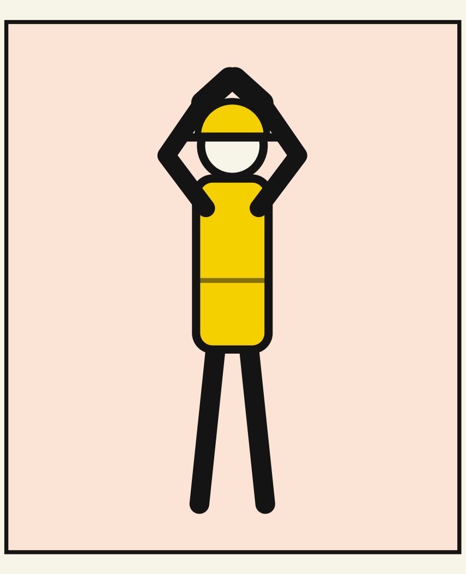
SITE SIGNAL

HEAT CASUALTY
Forearm across your brow, then point at the casualty. Bring ice and the medic.



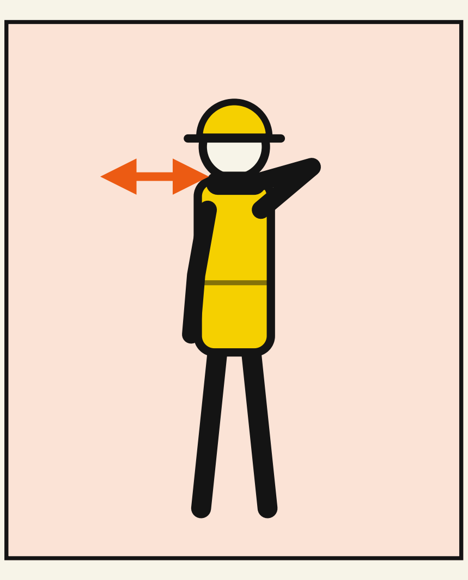
SITE SIGNAL

EVACUATE TO MUSTER
Whole arm straight, open hand, swung in big circles at your side like a windmill. Then point toward your muster point.



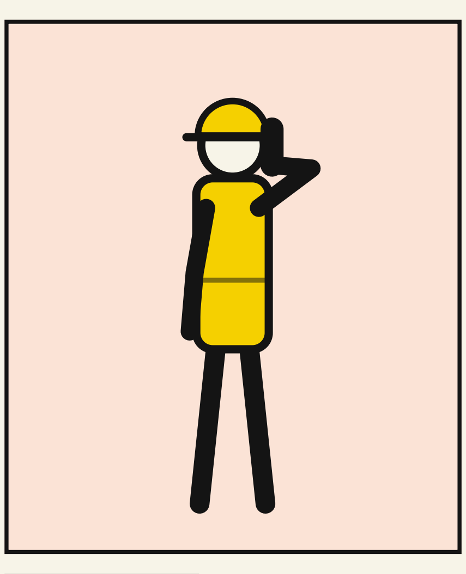
SITE SIGNAL

TAKE SHELTER
Both hands form a pointed roof above your head, fingertips touching, hands clear of the helmet. Then point to the nearest shelter.



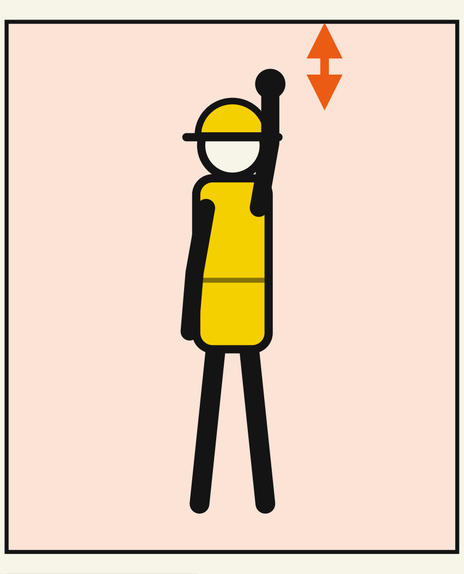
SITE SIGNAL

CAN'T BREATHE / AIR PROBLEM
Flat hand slicing across your throat. Respirator, cooling suit or air supply failing. Get them to clean air.



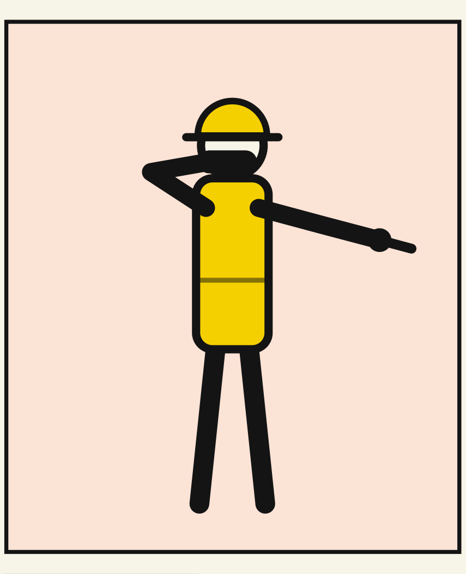
SITE SIGNAL

RADIO FAILURE
Open hand cupped behind your ear, held still: my radio is dead, or I can't hear you. Move closer, find a working radio or send a runner.



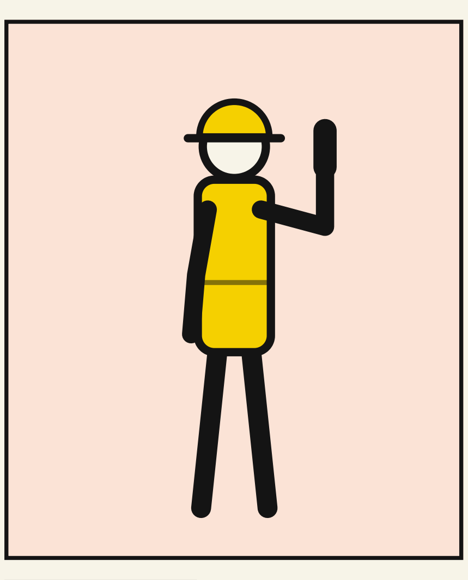
SITE SIGNAL

FIRE
Fist pumped straight up and down overhead. Then point at the fire and follow the fire rule below.



SITE SIGNAL

GAS / BAD AIR
Hand flat over your mouth and nose, then point upwind. Everyone moves upwind and out.



SITE SIGNAL

HOLD POSITION
Forearm up, flat hand facing the person. Stay where you are until waved on.

Site signals are for people, never for operators, except ALL STOP (EMERGENCY STOP). Near a lift, give EMERGENCY STOP first, then the site signal once the load is held. After any emergency signal, call it in on Ch 1 as soon as you can: "Emergency, emergency, emergency," type, location, number hurt, hazards, your call sign. The caller stays on Ch 1 until Hub Command or the medic releases them.

Fighting a fire (HS-09): fight it only if it's waste-bin size, your exit is behind you, and you have the right extinguisher. Not out in about 10 seconds? Leave and call Ch 1 "Fire". Never fight a polymer, power-cell or gas-fed fire. A polymer fire is a fire in bulk feedstock, a curing resin mass, a printed structure or panel stack, or a spray zone. A waste-bin-size fire in rags or scrap may be fought. Smother a rag can with its lid. A fire that's out: tell your foreman, who logs it on Ch 13.

STOP BEATS EVERYTHING

One arm out, palm down, swung forward and back = **STOP**. Both arms = **EMERGENCY STOP**, the site ALL STOP. Every operator obeys it from anyone: cranes, machines and hot work stop, and everyone holds position. A suspended load is landed only on the signal person's direction. The voice version is "ALL STOP, ALL STOP" on any channel. The siren version is **3 short blasts**. Cleared only by whoever called it, or by command.

Adapted from Old Earth standards: ASME B30.5 (Mobile and Locomotive Cranes) standard hand signals, OSHA 1926 Subpart CC Appendix A, OSHA 1926.1419 (signals), 1926.1420 (radio signals) and 1926.1428 (signal person qualification), common earthmoving spotter practice. Site emergency signals are Dunewright Colony Works site rules.

REF-HS · Dunewright Colony Works · Replace when faded